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Glove Based American Sign Language Interpretation Using Convolutional Neural Network and Data Glass

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***Abstract*—To live in a society, it is very important to communicate with each other. But this poses a grave problem for people with hearing disabilities. As they can converse using only sign language, it becomes very difficult for others who don't know the sign language to understand them. So, the purpose of this paper is to create an interpreter which can convert american sign language into the Engilsh language. Through convolutional neural network we were able to create such an interpreter which can interpret the american sign language.**

The Sign Language interpretation system was created using image processing, convolutional neural network and data glass. First, we created a database of 16500 samples (500 samples for each letter/gesture) for the basic sign language letters and gestures. Then using convolutional neural network, we were able to create a model which can distinguish between 33 hand gestures (26 English alphabets and 7 gestures).The images are taken through a raspberry-pi camera and then they are processed and matched against the model by a raspberry pi and the interpreted result is shown through a data glass.

***Index Terms*—American sign language, image processing, convolutional neural network, data glass.**

I. Introduction

Sign language is a system of non-verbal communication using visual gestures, symbols, facial expressions, body language etc. without the need of vocal communication. Any sign language has three main level of communication[1]. The first one is the finger spelling which means that for every alphabet in the language, there are corresponding signs allocated. This is the most basic form of sign language. Every word can be spelled by the proper use of the signs in sequence. The second level of communication is that there are some special signs which correspond to a whole word to speed up the whole communication process.The third one is the communication through the body language, eye movement, facial expression etc. But this form of communication is usually not preferred due to its misinterpretation and tardiness. So, among these three, the first two level of communication are the most used by the hearing disabled community for its speed and reliability[1].

So, in this paper we presented a system of interpreting american sign language. People with hearing disabilities will use sign language in front of a raspberry-pi camera and then the images extracted through the camera will be processed by the raspberry-pi and matched against the existing database and then the interpreted result will be shown by a data glass.

The rest of the paper is organized as follows: the section II contains the experimental results of previous researches in this area. Then in section III and IV, the neural network and convolutional neural network is explained in a simple way. The system architecture is explained in brief in the section V. Then section VI, VII, VIII contains the experimental results, conclusion and future scopes in this area.

II. Literature Survey

Dewinta Aryanie and Yaya Heryadi created a finger spelling recognition system using the K-Nearest Neighbour (KNN) classifier. Their research show that for K=3 their classifier achieves the higher accuracy (99.8%) with the full dimensional feature [2]. But their accuracy drops to 28.6% when the pattern is represented with PCA (Principal Component Analysis) reduced dimensional feature [2]. To explain the variability of the original dataset, the principal components were selected based on its Eigenvalues [2].

Kshitij Bantupalli and Ying Xie used a CNN (Convolutional Neural Network) model named Inception to extract spatial features from the video stream for Sign Language Recognition [3]. Then, the temporal features were extracted by using the Long Short-Term Memory and Recurrent Neural Network model [3]. According to their research, their model had an accuracy of 99% after the training stage.

Haar Cascade Classifier was used by Kanchan Dabre and Surekha Dholay to classify and train each hand gestures. Then a speech synthesizer was used to convert the displayed text into speech [1]. Their research show that they were able to attain an average accuracy of 92.68% through the Haar Cascade features [1].

Tülay Karayilan and Özkan Kiliç applied the backpropagation algorithm to create the neural network in order to classify the hand gestures. They were able to acquire accuracy of 75%-85% with two classifiers [4]. One of them was used for raw features and the other one for histogram features [4]. For dataset, they used the Marcel Static Hand Posture [4].

Xinyun Jiang and Wasim Ahmad also created a neural network using the Support Vector Machine algorithm to classify and train the hand gestures. Their experimental result show that the system was able to distinguish 5 alphabets (B, D, F, L, U) with a success rate of about 99.4% [5].

III. Neural Network

An artificial neural network consists of artificial neurons. The figure 1, gives a simplified view how an artificial neuron works. Here, the X1, X2, X3 are the input data of a neuron. The inputs then are multiplied by unique weights W1, W2, W3 respectively and then summed together. The summation is passed through a threshold function.Theoretically, it should be a step function. When the sum value is more than the threshold value, the output is 1 otherwise it

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is 0. But a step function has only the value 1 and 0 which is not ideal for the real world. So, instead a sigmoid function is used. If a sigmoid function is used, it is not called a threshold function anymore. Rather, it is called an activation function. So, the whole function can be written as y= f(XW). These values of y's are then fed into the next neuron.

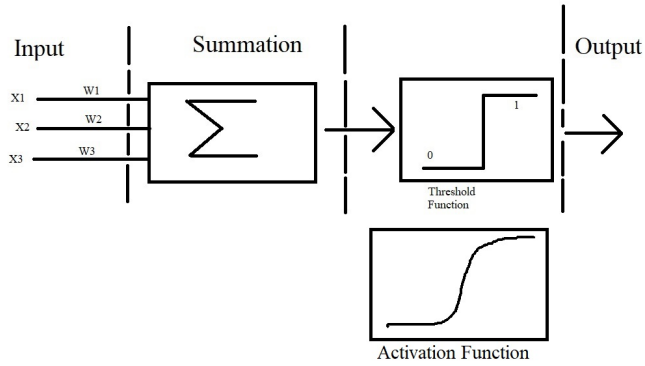


Fig. 1. Operation of an artificial neuron

The figure 2, gives a simplistic view of how a neural network works. Again, the inputs are X1, X2 and X3. Consider the little circles as artificial neurons. The inputs are connected to the each of the neurons and each of the connections has a unique weight associated with it. This is the 1st hidden layer of neuron and each of the neurons are connected to next layer of neurons (2nd hidden layer) with another unique set of weights and then they are connected to output layer and again with each unique weight [6].

This is an example of a deep neural network. Because if a model has one hidden layer, it is termed as a regular neural network but if it has more than one, it is called a deep neural network.

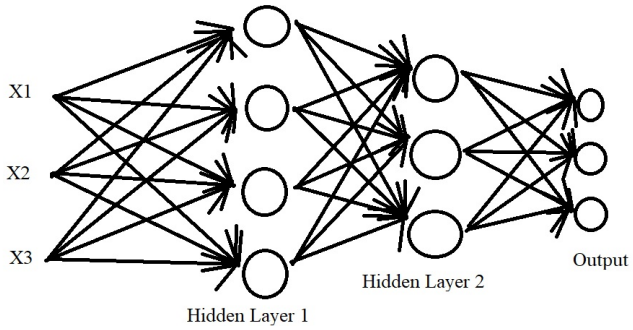


Fig. 2. Neural network

IV. Convolutional Neural Network

Convolutional Neural Network is state of the art neural network model for recognizing what's in an image or what an image is. Basic operations of a convolutional neural network are shown in the block diagram.

A. Convolution

Convolution is a form of mapping or creating a feature map from a primary dataset. Basically, there is going to be a moving region over the entire image either looking for something or trying to create a feature map with a region of pixels (n*n) in order to classify the image [7].

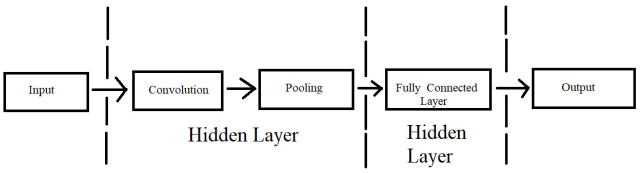


Fig. 3. Convolutional neural network block diagram

Now, let's consider the figure 4(a) is an image of a tree and has been broken down to pixels (square boxes). For a convolution of 3*3, a region of 3*3 is going to be created and it will move over the entire image either for searching or classifying it until no pixels are left and each of the features will be assigned a value.

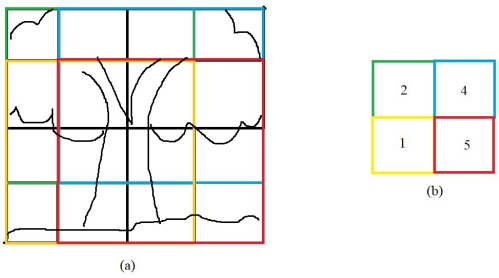


Fig. 4. (a) An image of a tree broken down in pixels (b) Feature map extracted from convolution

B. Pooling

Let's assume that after convolution we get the figure 5(a). Now, this is going to be the new feature set and for a pooling of 3*3, a region of 3*3 is going to move over the feature set and pool the maximum value from that region.

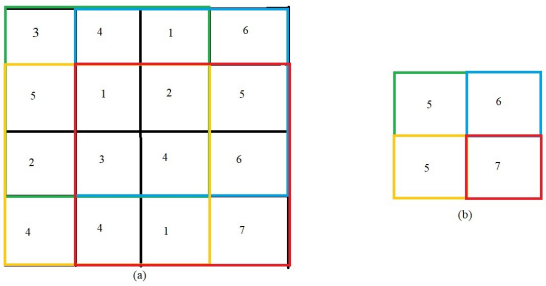


Fig. 5. (a) Feature map after convolution (b) Pooling

Our convolutional neural network model consists of one input layer, 7 hidden layers, 1 fully connected layer and 1 output layer. For optimization, we used Adam optimizer and we trained the model for 10 epochs.

V. System Architecture

The total system architecture is given in the figure 6. As shown in the figure, the entire system works in three stages. A general description of these stages is given below:

A. Modeling Stage

In this stage, the model was created using the processed images. So, at first the images were extracted frame by frame from a video feed.

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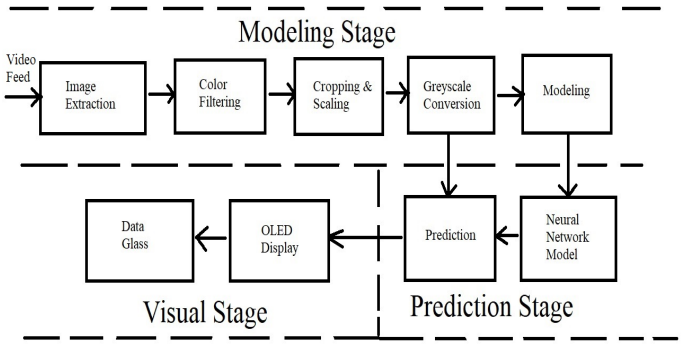


Fig. 6. System architecture

Then they were passed through various image processing techniques such as color filtering, cropping, scaling, noise removal, greyscale conversion etc. And lastly, through convolutional neural network we were able to create the neural network model for sign language.

1) Color Filtering: As this is a glove based sign language interpretation system, through color filtration we segmented the hand region from the rest of the environment. This way, we first created our database of 16500 samples. Later in the prediction stage, through color filtration, the hand region is again segmented and then matched against the existing database.

2) Cropping And Scaling: Although the unwanted sections were filtered out, the image size was still as same as before filtering. To create a neural network model with such large images can be quite a lengthy process and also unnecessary. Because, neural network model works through pixel by pixel. It is not dependent on the image sizes. Rather, the smaller the image size, the faster the neural network is. So, the sample images were cropped and scaled to 480*360 pixels.

3) Removal of Noise: Even after the color filtering and cropping, there were still some noise left on the images. These noises can cause severe complication during the prediction stage of the system if they are left on the images. So, for better performance with the help of Gaussian filter unwanted noise was removed from the extracted images.

4) Greyscale Conversion: As most of the OpenCV functions need greyscale images, the RGB images were converted to Greyscale images. But this process can be skipped because if the images are not in greyscale, the OpenCV will do it by default under the hood.

5) Creation Of The Model: After the images have been processed, they are used as the database for the neural network. Then using convolutional neural network, the model was created. Although the database was created by the raspberry pi, the modeling was not possible through it. As the raspberry pi lacks in RAM and processing capability, the model was created through a Laptop with high configuration.

B. Prediction Stage

In this stage, the images are again extracted from the video feed of a raspberry pi camera and they go through the same image processing techniques as explained before. But this time the model is already created. So, all we really need to do is pass the processed image through the neural network and match it against the database and predict the signs. The neural network matches the processed images against the all the images contained in its database and returns the probability of the processed image being the same image in the database. The image with highest matching probability is given as the output.

C. Visual Stage

The real task of the next stage was to make the output of the neural network understandable to humans. Somehow, the output had to be cast on a screen for human eyes to see. This is where the data glasses came to our rescue. As the matched images corresponded to each unique letters or words, through raspberry pi, the corresponding letters or words were displayed on the screen of a data glass. The main components of data glass is divided into 3 parts.

1) OLED Display: This is the first part that is connected with raspberry pi and the raspberry pi outputs the letters/words on it.

2) Mirror And Lens: The Second part is the mirror which passes the light from OLED to data glass. To focus perfectly, we used a lens which passes the light from mirror to data glass.

3) Data Glass: The main portion is the data glass on which the letters/words are shown. Data glass is movable so that it can be adjusted for perfect focusing. The whole system works like a periscope.

VI. Experimental Result

The figure 9 and 10 show accuracy and loss of the trained model. From the graph we can see that we were able to attain an accuracy of 91.25% and loss was reduced to 0.40669 which is far better than that can be achieved through any other algorithm.

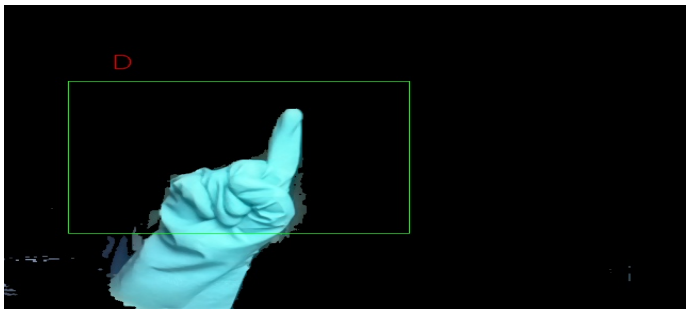


Fig. 7. Letter ”D” interpreted

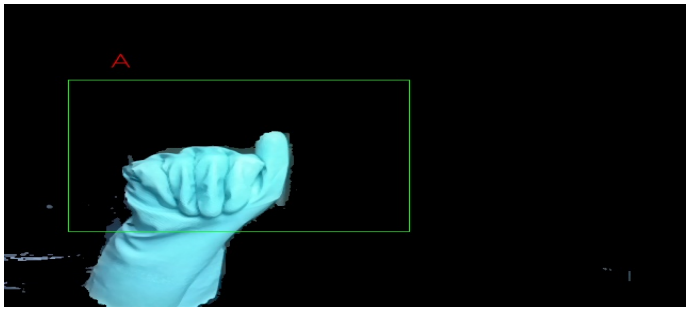


Fig. 8. Letter ”A” interpreted

This model is able to recognise 33 letters and gestures (26 alphabets and 7 gestures). At first, we had trained the model with 5 hidden layers and for 5 epochs and with a dataset of 9900 samples (300 samples each). But we were able to achieve an accuracy level of only 74.37%. So, we increased the no. of samples, hidden layers and epochs. Then we were able to achieve an accuracy of 91.25%.

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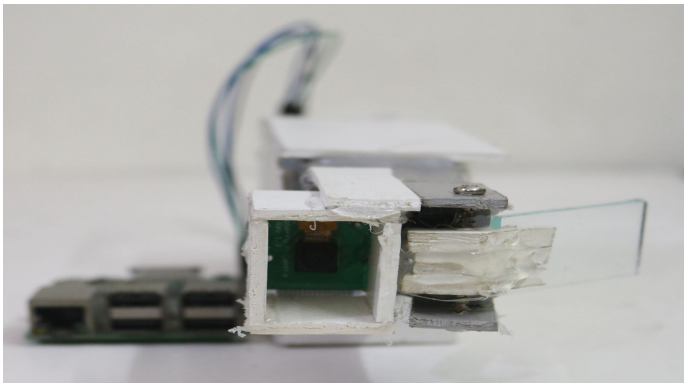


Fig. 9. Raspberry pi and data glass interface

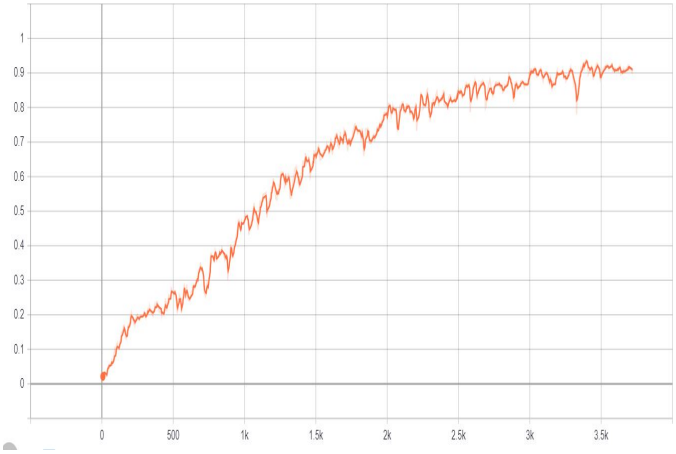


Fig. 10. Accuracy vs Training steps

VII. Conclusion

In this paper, the interpretation of American Sign language using convolutional neural network is presented.This model is well known for its faster data classification and convergence than its predecessors such as backpropagation algorithm, support vector machine, k-nearest neighbor method.

Dewinta Aryanie and Yaya Heryadi were able to achieve an accuracy of 99.8% through the K-Nearest Neighbour Method [2]. But this model actually requires a long and tedious operation. So, with larger dataset like american sign language this model is very slow compared to the convolutional neural network.

Xinyun Jiang and Wasim Ahmad's neural network model using Support Vector Machine also shows an accuracy of 99.4% [5]. But their model was able to distinguish only 5 differnt alphabets whereas with convolutional neural network we were able to distinguish all 26 alphabets of English language and 7 other gestures.

Another effective feature of this model is that it is also supported by a raspberry-pi operating software. This is the key feature which makes it possible to interpret the sign language through data glass. Although our overall accuracy is 91.25%, our model is faster in sign recognition than any other previous model. To gain this speed advantage, we had to compromise the accuracy level a bit.

VIII. Future Scope

So far, we have used a raspberry-pi 3 to process the images and match it against the neural network model and predict the signs.

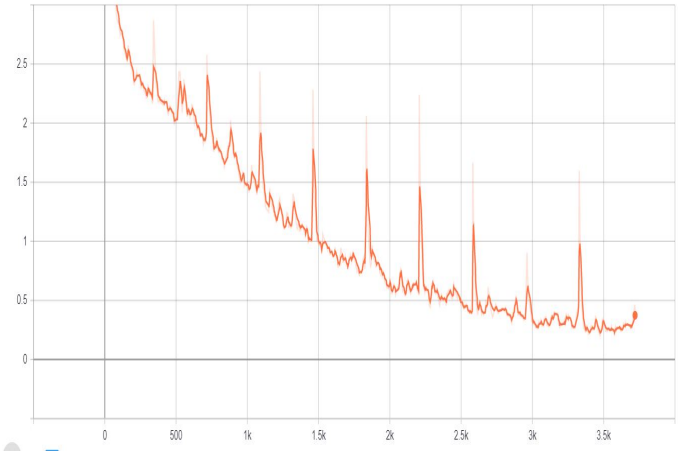


Fig. 11. Loss vs Training steps

But it is bulky and heavy. As the pi camera has to be connected to both raspberry pi and data glass at the same time, it can be pretty troublesome to keep them in one place. So, in the future we are planning to use raspberry pi zero w in this case. Because although it is pretty small and lightweight, it has all the required processing capabilities. As neural network for data modeling is relatively new, the necessary modules for the data classification and recognition for raspberry pi zero w still have some bugs. But the raspberry pi community is working relentlessly to fix the issues. So, with raspberry-pi zero, this project can be applicable in the real world.

Due to lack of processing power of raspberry-pi we had to implement a glove based interpretation system. This system identifies the glove in front of the camera and segments it from rest of the environment and then matches against the existing database. The existing raspberry-pi models are unable to provide essential computational power required for skin filtration and segmentation. Lagging and heating of the raspberry-pi was noticed. So, with new model of raspberry-pi we hope that we will be able to provide a interpretation system with skin filtration and segmentation.

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